

Francesco Valentini - Curriculum Vitæ

Personal Details

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Address: Università di Pisa,
Dipartimento di Economia e Management,
Via Cosimo Ridolfi 10, 56124, Pisa, Italy

Current Position

Assistant Professor of Econometrics (ECON-05/A)
Department of Economics and Management, University of Pisa, Pisa, Italy
Starting date: October 10, 2024

Research Interests

Panel Data, Fixed Effects, Discrete outcomes, Network Econometrics, Applied Microeconometrics,
Applied Time Series Analysis

Past Positions

- Junior Assistant Professor of Statistics (SECS-S/01)
Department of Economic and Social Sciences, Marche Polytechnic University, Ancona, Italy
January 1, 2023 - October 9, 2024
- Postdoctoral Fellow in Econometrics (SECS-P/05) on the Project:
“Panel data models and machine learning methods for the Green Innovators’ performance evaluation”
Tutor: Prof. Claudia Pigini
Department of Economic and Social Sciences, Marche Polytechnic University, Ancona, Italy
September 1, 2022 - December 31, 2022
- Postdoctoral Fellow in Econometrics (SECS-P/05)
Project: “Conditional inference for discrete panel data models”
Tutor: Prof. Claudia Pigini
Department of Economic and Social Sciences, Marche Polytechnic University, Ancona, Italy
June 1, 2020 - May 31, 2022

Education

19/03/2020 - Ph.D. in Economics, Marche Polytechnic University
Doctoral thesis: “Three Essays on the Conditional Inference Approach for Binary Panel Data Models”
Supervisors: Prof. Francesco Bartolucci and Prof. Claudia Pigini

24/10/2015 - Master’s Degree in Economics and Finance, Marche Polytechnic University,
110/110 magna cum laude

Publications and Research Activities

– *Peer-Reviewed Articles*

Forthcoming - Claudia Pigini, Alessandro Sterlacchini and Francesco Valentini. “Persistence of R&D intensities in the world’s top investors in R&D”, *Economics of Innovation and New Technology*

2025 - Michele Fratianni, Federico Giri, Riccardo Lucchetti and Francesco Valentini. “Monetization and the Fiscal Multiplier”, *Oxford Bulletin of Economics and Statistics*,
<https://doi.org/10.1111/obes.12658>

2024 - Francesco Bartolucci, Claudia Pigini and Francesco Valentini. “MCMC Conditional Maximum Likelihood for the two-way fixed-effects logit”, *Econometric Reviews*,
<https://doi.org/10.1080/07474938.2024.2339145>

2024 - Riccardo Lucchetti and Francesco Valentini. “Linear models with time-varying parameters in gretl: a comparison of different approaches”, *Computational Statistics*,
<https://doi.org/10.1007/s00180-023-01452-3>

2023 - Riccardo Lucchetti and Francesco Valentini. “Kernel-based Time-Varying IV estimation: handle with care”, *Empirical Economics*, <https://doi.org/10.1007/s00181-023-02450-6>

2022 - Francesco Bartolucci, Claudia Pigini and Francesco Valentini. “Testing for state dependence in the fixed-effects ordered logit model”, *Economics Letters*,
<https://doi.org/10.1016/j.econlet.2022.110964>

2022 - Francesco Bartolucci, Claudia Pigini and Francesco Valentini. “Conditional inference and bias reduction for partial effects estimation of fixed-effects logit models”, *Empirical Economics*,
<https://doi.org/10.1007/s00181-022-02313-6>

2021 - Francesco Bartolucci, Francesco Valentini and Claudia Pigini. “Recursive computation of the conditional probability function of the quadratic exponential model for binary panel data”, *Computational Economics*, <https://doi.org/10.1007/s10614-021-10218-2>

2021 - Chiara Casoli, Luca Pedini and Francesco Valentini. “Do Covid-19 mobility restrictions affect economic uncertainty in Italy? Evidence from a SVAR approach”, *Economics Bulletin*

– *Conference Proceedings*

2019 - Chiara Casoli, Luca Pedini and Francesco Valentini. “Spatial models in gretl: the SPM package” in *GRETl 2019. Proceedings of the International Conference on the Gnu Regression, Econometrics and Time-series Library*, DOI:10.6093/978-88-6887-057-7

– *Books*

2023 - Francesco Valentini, Claudia Pigini and Francesco Bartolucci. “Advances in maximum likelihood estimation of fixed-effects binary panel data models” in *Trends and Challenges for Categorical Data Analysis*, edited by Maria Kateri and Irini Moustaki, Springer,
https://doi.org/10.1007/978-3-031-31186-4_9

– *Work in Progress*

- “Grouped fixed effects regularization for binary choice models” (joint with C. Pignini and A. Pionati)
- “Specification testing with Grouped Fixed Effects” (joint with C. Pignini and A. Pionati)
- “Testing for dependence in pairs of binary outcomes by an approximating logit model”

– *Software Components*

- cquad: Conditional Maximum Likelihood for Quadratic Exponential Models for Binary Panel Data.
R package
Francesco Bartolucci, Claudia Pignini and Francesco Valentini. 2020
<https://cran.r-project.org/package=cquad>
- cquadr: run the cquad R package in Stata.
STATA package
Francesco Bartolucci, Claudia Pignini and Francesco Valentini. 2020
<https://ideas.repec.org/c/boc/bocode/s458852.html> and <https://github.com/fravale/cquadr>
- FLS: Flexible Least Squares.
gretl function package
Riccardo “Jack” Lucchetti and Francesco Valentini. 2023
https://gretl.sourceforge.net/current_fnfiles/FLS.zip
- ketvals: Kernel-based Time-Varying Least Squares.
gretl Function Package
Riccardo “Jack” Lucchetti and Francesco Valentini. 2021
https://gretl.sourceforge.net/current_fnfiles/ketvals.zip
- spm: Spatial Regression Models.
gretl Function Package
Chiara Casoli, Luca Pedini and Francesco Valentini. 2019
https://gretl.sourceforge.net/current_fnfiles/spm.zip

– *Research projects*

- Member of the research project:
Next Generation EU. Project CUP: Project Code ECS00000041, CUP I33C22001330007;
Title: Innovation, digitalization and sustainability for the diffused economy in Central Italy – VITALITY
Period: January 1, 2023 - October 9, 2024

Refereeing Activity

Econometrics, Journal of Economic Behavior & Organization, Psychometrika (Theory and Methods),
SN Business & Economics, Statistical Methods and Applications

Talks/Presentations

2025 - Invited speaker at “Special Seminars InsIDE Lab”, University of Insubria, Varese, Italy.
Presentation: “Grouped fixed effects regularization for binary choice models”

2025 - 11th Italian Congress of Econometrics and Empirical Economics, Palermo, Italy.
Presentation: *"Specification testing with Grouped Fixed Effects"*

2023 - The 28th International Panel Data Conference, Amsterdam, Netherlands.
Presentation: *"Specification testing with Grouped Fixed Effects"*

2023 - 8th gretl Conference, Gdańsk, Poland.
Presentation: *"Linear models with time-varying parameters in gretl: comparing different approaches"*

2023 - 10th Italian Congress of Econometrics and Empirical Economics, Cagliari, Italy.
Presentation: *"Monetization and the fiscal multiplier"*

2022 - The 27th International Panel Data Conference, Bertinoro, Italy.
Presentation: *"MCMC Conditional Maximum Likelihood for the two-way fixed-effects logit"*

2022 - Challenges for Categorical Data Analysis 2022, Perugia, Italy.
Presentation: *"Dynamic Ordered Logit with Fixed-Effects: testing for state dependence"*

2022 - 3rd Italian Workshop of Econometrics and Empirical Economics, Rimini, Italy.
Presentation: *"MCMC Conditional Maximum Likelihood for the two-way fixed-effects logit"*

2021 - 9th Italian Congress of Econometrics and Empirical Economics, Cagliari, Italy [Web].
Presentation: *"Partial effects estimation for fixed-effects logit panel data models"*

2019 - 7th SIdE Workshop for PhD students in Econometrics and Empirical Economics, Bertinoro, Italy.
Presentation: *"A conditional approach to testing for endogenous sample and self-selection in fixed-effects logit panel data models"*

2019 - 6th gretl Conference, Naples, Italy.
Presentation: *"Spatial models in gretl: the SPM package"*

Teaching

2025/2026 - Member of the PhD Board of "PhD Programme in Economics" of the Tuscan Universities, University of Siena

2025~2026- Lecturer of "Econometrics for Financial Markets",
Master in Risk Management, University of Pisa

2024/2025 ~ 2025/2026 - Lecturer of "Macroeconometrics",
Master's Degree in Economics, University of Pisa

2024/2025 ~ 2025/2026 - Lecturer of "Advanced Econometrics",
Master's Degree in Economics, University of Pisa

2024/25 - Lecturer of "Statistics for technological and experimental research",
Bachelor's Degree in Cardiac and Cardiovascular Perfusion techniques,
Faculty of Medicine and Surgery, Marche Polytechnic University

2023/2024 ~ 2024/25 - Lecturer of "Statistics for technological and experimental research",
Bachelor's Degree in Biomedical Laboratory Techniques, Faculty of Medicine and Surgery,
Marche Polytechnic University

2023/24 - Lecturer of "Econometric methods in practice"
(coordinator: Prof. Claudia Pigini), Ph.D. in Economics, Marche Polytechnic University

2019/20 ~ 2023/24 - Lecturer of “Microeconometrics in practice”
(coordinator: Prof. Matteo Picchio), Ph.D. in Economics, Marche Polytechnic University

2019/20 ~ 2022/23 - Lecturer of “Time Series Econometrics in practice”
(coordinator: Prof. Giulio Palomba), Ph.D. in Economics, Marche Polytechnic University

2020/21 ~ 2022/2023 - Teaching assistant: “Economia Politica I” (Microeconomics),
Bachelor’s Degree in Economics and Business, Marche Polytechnic University

2020 ~ 2021 - Teaching assistant: “Practice sessions in R”
(coordinators: Prof. Riccardo Lucchetti and Prof. Maria Cristina Recchioni)
Specialization Course in Data Science for Enterprises, Istituto Adriano Olivetti, Ancona, Italy

2019/20 - Associate Lecturer (Docente a Contratto) of “Economic and Financial Time Series Analysis”,
Master’s Degree in Finance and Quantitative Methods for Economics,
Department of Economics, University of Perugia, Perugia, Italy

2016/17 ~ 2018/19 - Tutor of “Economia Politica I” (Microeconomics),
Bachelor’s Degree in Economics and Business, Marche Polytechnic University

Memberships

2017-2019, 2021-2026 - SIIE-IEA, *Italian Econometric Association*

Other Experiences

2025 - The 30th International Panel Data Conference, Montpellier, France

2024 - The 29th International Panel Data Conference, Orléans, France

02/2019 ~ 07/2019 - Visiting Ph.D. student at University of Perugia, courses attendance:

“Gaussian Mixture Models for Model Based Clustering and Classification”, Prof. Luca Scrucca

“Introduction to Latent Variable Models”, Prof. Francesco Bartolucci

2019 - 8th Italian Congress of Econometrics and Empirical Economics, Lecce, Italy

2018 - 1st Italian Workshop of Econometrics and Empirical Economics, Milan, Italy

Languages

Italian: Mother tongue

English: Upper Intermediate, Cambridge First Certificate in English (B2 Level)

German: Basic, basic words and phrases only

Computer Skills

Operating Systems: GNU/Linux, MacOS, Microsoft Windows

Text Editing and Office Skills: $\text{\LaTeX}$, LibreOffice, Microsoft Office

Software: R, gretl, STATA, MATLAB

Last update: November 19, 2025